

$$C_p = \frac{P - P_\infty}{q_\infty}$$

$$q_\infty = \frac{1}{2} \rho_\infty V_\infty^2 = \frac{1}{2} \rho_\infty M_\infty^2 \gamma R T_\infty$$

$$\rho_\infty R T_\infty = P_\infty$$

$$\Rightarrow q_\infty = \frac{1}{2} \gamma P_\infty M_\infty^2$$

$$C_p = \frac{p - p_\infty}{q_\infty}$$

$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 = \frac{1}{2} \frac{\gamma p_\infty}{\gamma p_\infty} \rho_\infty U_\infty^2$$

$$= \frac{\gamma p_\infty U_\infty^2}{2 a_\infty^2} = \frac{\gamma p_\infty M_\infty^2}{2} = q_\infty$$

$$\Rightarrow C_p = \frac{p - p_\infty}{\frac{\gamma p_\infty M_\infty^2}{2}} \Rightarrow \boxed{C_p = \frac{2}{\gamma M_\infty^2} \left(\frac{p}{p_\infty} - 1 \right)}$$

①/② ~~①/②~~

[∞] for 1 and 2 conditions: ^{nothing}

$$C_p = \frac{2}{\gamma M_1^2} \left(\frac{p_2}{p_1} - 1 \right)$$

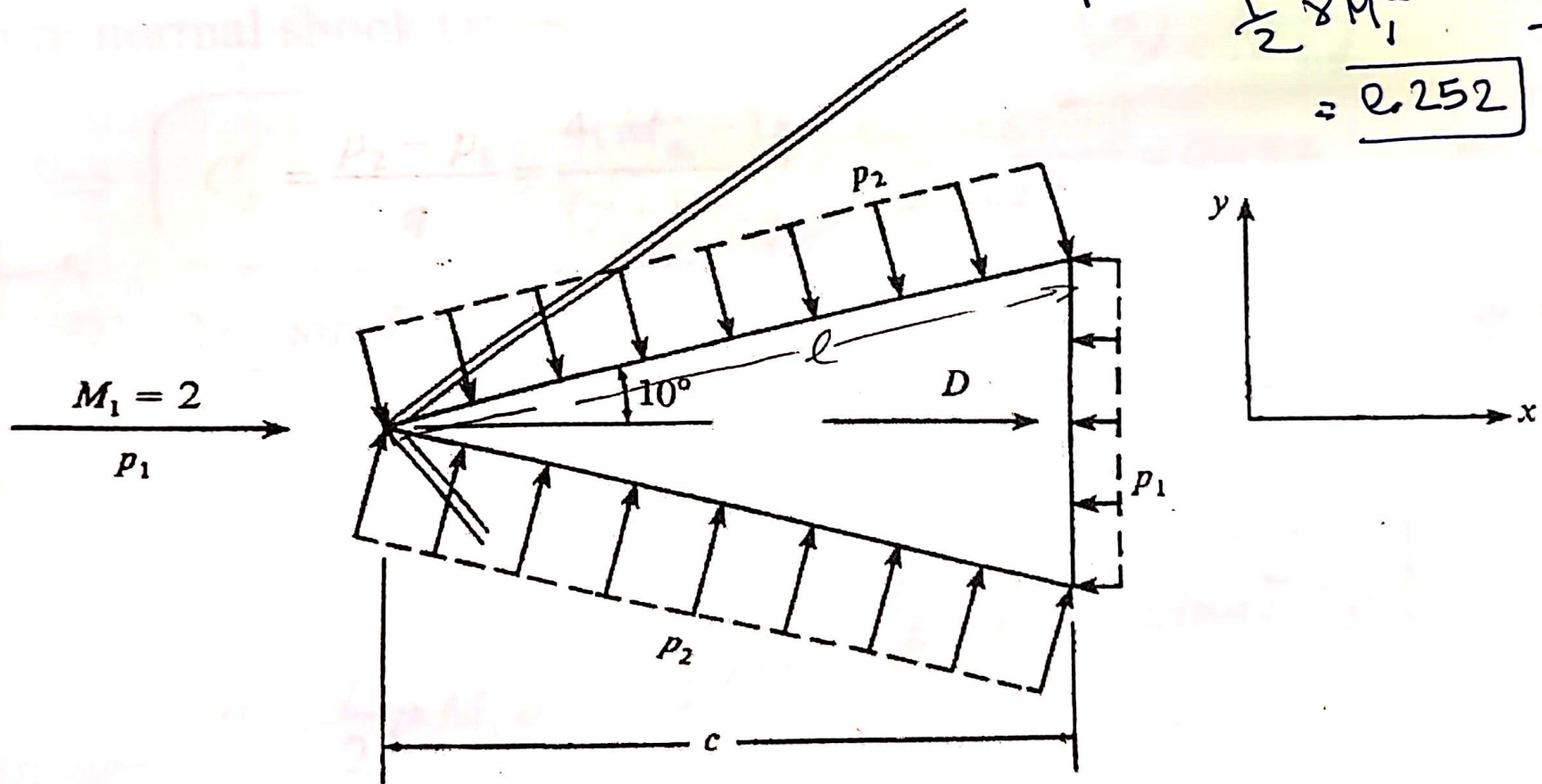
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for incompressible flow:

$$C_p = 1 - \left(\frac{V}{V_\infty} \right)^2 \quad \left(\text{using Bernoulli's eq.} \right)$$

Consider a 10° wedge, $M_1=2$ find C_p , C_d

$$C_p = \frac{P_2 - P_1}{\frac{1}{2} \gamma P_1 M_1^2}$$

$$C_p = \frac{\frac{P_2}{P_1} - 1}{\frac{1}{2} \gamma M_1^2} = \frac{1.707 - 1}{\frac{1}{2} \times 1.4 \times 2^2} = \underline{0.252}$$



$M_1=2$, $\delta=10^\circ$ from $\theta-\delta-M$ diagram $\theta = 39.314^\circ$

$$C_p = \frac{P_2 - P_1}{q_\infty} = \frac{\frac{P_2}{P_1} - 1}{\frac{1}{2} \gamma M_\infty^2}$$

$$\Rightarrow \underline{\underline{M_{n_1} = M_1 \sin \theta = 2 \times \sin 39.314 = 1.267}}$$

from normal shock tables:

$$M_{n_1} = 1.267 \Rightarrow \boxed{\frac{P_2}{P_1} = 1.707}$$



$$\Rightarrow C_p = \frac{P_2 - P_1}{q} = \frac{4(M_{n_1}^2 - 1)}{(\gamma + 1)M_1^2} = \frac{4(1.267^2 - 1)}{2.4 \times 2^2} = 0.252$$

same answer

also $C_p = \frac{2}{\gamma M_1^2} \left(\frac{P_2}{P_1} - 1 \right)$ (or use gauge pressure)

$D' = 2p_2 l \sin \delta - 2p_1 l \sin \delta = 2(\sin \delta) l (P_2 - P_1)$
 / 2 oblique sides vertical side
 - horizontal component

drag/unit span = $2c \tan \delta (P_2 - P_1)$

$$C_d = \frac{D'}{q_1 c} = \frac{D'}{\frac{\gamma}{2} P_1 M_1^2 c} = \frac{4 \tan \delta}{\gamma M_1^2} \left(\frac{P_2}{P_1} - 1 \right) = 2 \tan \delta C_p$$

$$= 2 \times \tan 10^\circ \times 0.252$$

$$= \boxed{0.184}$$

note $C_d \neq 0$; oblique shock creates drag

$$\frac{P_2}{P_1} = \left[1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right] P_2$$

M_{n1} ①
②

$$P_2 - P_1 = \left[\left(1 + \frac{2\gamma}{\gamma+1} (M_1^2 - 1) \right) P_1 \right] - P_1 = \frac{2\gamma}{\gamma+1} (M_1^2 - 1) P_1$$

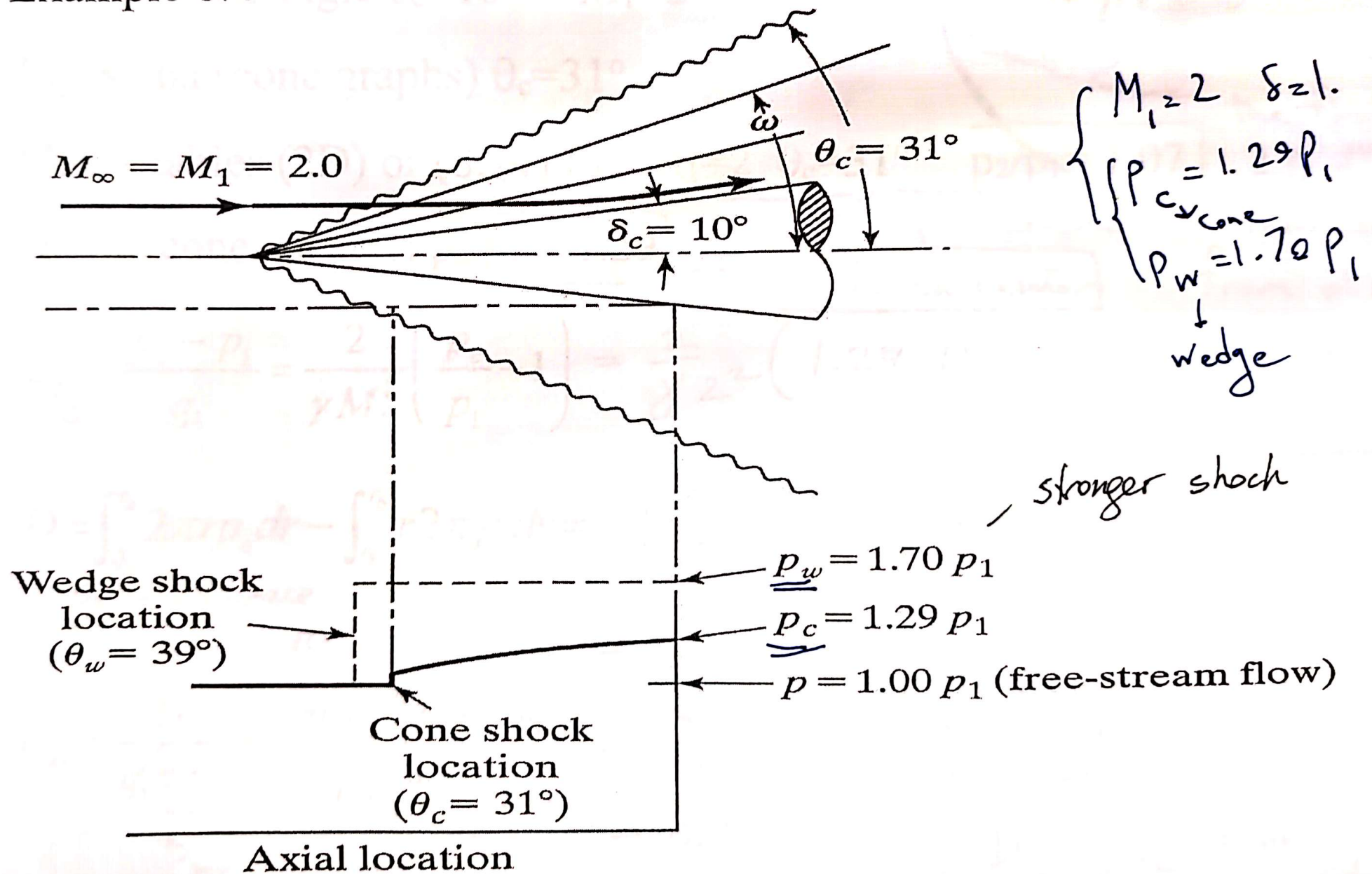
$$C_p = \frac{P_2 - P_1}{\frac{1}{2} \gamma M_1^2 P_1} = \frac{\frac{2\gamma}{\gamma+1} (M_1^2 - 1) P_1}{\frac{1}{2} \gamma M_1^2 P_1} = \frac{4 (M_1^2 - 1)}{(\gamma+1) M_1^2}$$

$$C_p = \frac{P_2 - P_1}{\frac{1}{2} \gamma M_1^2 P_1} = \frac{\frac{2\gamma}{\gamma+1} (M_1^2 - 1) P_1}{\frac{1}{2} \gamma M_1^2 P_1} = \frac{4 (M_1^2 - 1)}{M_1^2 (\gamma+1)}$$

$$C_p = \frac{4 (M_1^2 - 1)}{M_1^2 (\gamma+1)}$$

$$M_n = M_1 \sin \theta$$

Example 8.4



Semi-vertex angle $\delta_c = 10^\circ$ $M_1 = 2$

Fig 8.16a (cone graphs) $\theta_c = 31^\circ$

Shock tables (2D) or (8.74) for $M_1 = 2$ $\theta_c = 31^\circ$ $p_2/p_1 = 1.07$

On the cone surface: $p_c/p_1 = 1.29$

"oblique shock"

conical calculator

Just after
the shock
the flow is
2D flow can
read data
from oblique
shock relations

$$C_{p_c} = \frac{p_c - p_1}{q_1} = \frac{2}{\gamma M_1^2} \left(\frac{p_c}{p_1} - 1 \right) = \frac{2}{\gamma 2^2} (1.29 - 1) = 0.104$$

$$D = \int_0^{r_b} 2\pi r p_c dr - \int_0^{r_b} r 2\pi p_1 dr = \int_0^{r_b} 2\pi r (p_c - p_1) dr = \pi (p_c - p_1) r_b^2$$

oblique surface base
horizontal force

$$C_D = \frac{D}{q_1 (\pi r_b^2)} = \frac{\pi r_b^2 (p_c - p_1)}{\pi r_b^2 q_1} = C_p = 0.104$$

S_{base}

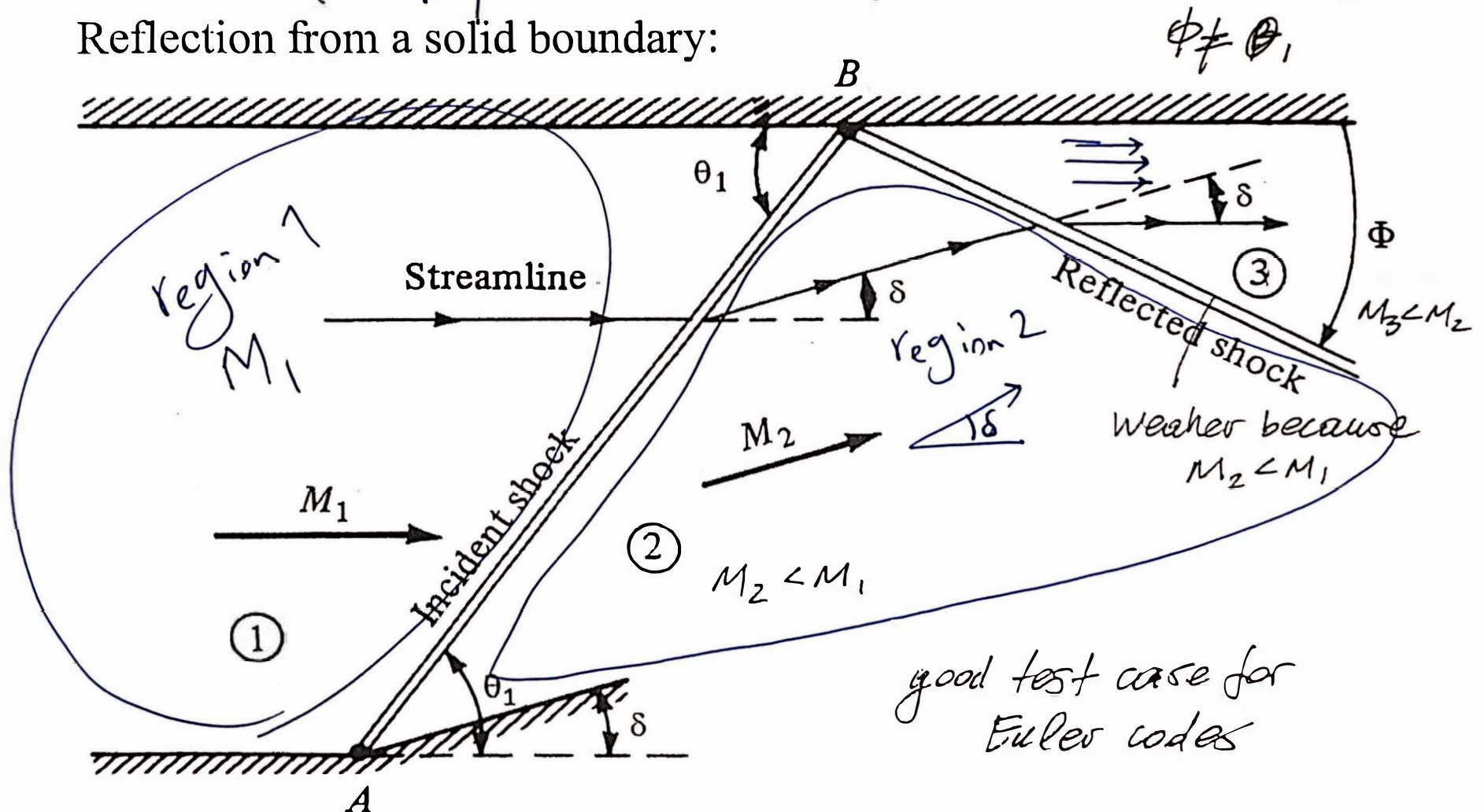
$$(C_D = C_p)$$

$$D_c = \int_0^{r_b} p_c 2\pi r dr - \int_0^{r_b} p_1 2\pi r dr$$

Second shock $\left\{ \begin{array}{l} M_2 \text{ read} \\ \delta_2 = \delta_1 \end{array} \right\} \rightarrow \theta$ First shock $\left\{ \begin{array}{l} M_1 \\ \delta_1 \end{array} \right\} \rightarrow \text{read } \underline{\theta}$

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Reflection from a solid boundary:



2 oblique shocks sequentially; second shock is weaker $\phi \neq \theta_1$
(because it starts at a lower $M \#$)

from $M_1, \delta \rightarrow \theta_1, M_2$; $M_2, \delta \rightarrow M_3, \theta_2$ $\phi = \theta_2 - \delta$